

ATP Problem Set 1

Problem 1.

a) Create an example similar to the ones from lecture 1 that illustrates the use of both the matching and the revenue recognition principle. Create two mini income statements (revenue, costs, profit) one on a cash basis and one on an accrual basis.

JLim opens a supercar rental business in Los Angeles. He acquired two Lamborghinis for 600,000 USD each in Year 1. He signs a two-year contract with Logan Paul, accepting a per-car payment of 170,000 USD to cover the garage, maintenance, and opportunity cost of not renting the car daily for two years. Assume the two cars have a useful life of ten years.

Under the revenue recognition principle, the revenue from Logan Paul's contract should be recognized over the contract period as the service is provided. So, the 170,000 USD that JLim received upfront will not be factored in at once and should be recognized evenly as 85,000 USD revenue per car per year.

Under the matching principle, the cost of each Lamborghini should not appear upfront as a negative 600,000 USD. Costs associated with delivering a service must be recorded during the same period as revenue generated from that product or service. Hence, if the useful life of each car is 10 years (2 years used by Logan Paul and 8 years rented to the public), then the per-year cost of each car should be stated as 60,000 USD. The 60,000 USD annual depreciation expense is matched against the rental revenue earned each year.

Cash Basis

Revenue (Year 1)	340,000 USD	Revenue (Year 2)	0
Costs (Year 1)	1,200,000 USD	Costs (Year 2)	0
Profit (Year 1)	-860,000 USD	Profit (Year 2)	0

Accrual Basis

Revenue (Year 1)	170,000 USD	Revenue (Year 2)	170,000 USD
Costs (Year 1)	120,000 USD	Costs (Year 2)	120,000 USD
Profit (Year 1)	50,000 USD	Profit (Year 2)	50,000 USD

Why are we making a profit under Accrual, but a loss under Cash? Under cash we record everything when cash moves, not when value is created. But under accrual, we spread our cost across the useful life of the car (ten years), and recognize the revenue from Logan Paul's contract across the two years it holds.

b) In a short paragraph explain why an investor would be interested in both the cash and accrual income statements.

An investor might be interested in the cash income statement because it might give them a clue about predicting future debt obligations. Especially in my example, JLim spends a ton of money up-front, which may require taking a loan which is important to understand because interest decreases future profits. Also, the cash income statement shows the company's liquidity when it is published. After buying the lambos JLim would be less liquid, which may inform some investment decisions. An investor might also be interested in the accrual income statement because it shows the investor the long term profitability of the business' capital allocation and its true performance. It shows how effective cash is being spent; if we match capital allocation to the revenue associated with it, we can effectively find the business' sustainable margin.

Problem 2. Fill in the income statement on the next page for the year ending December 2023 based on the following information

1. You enter the lemonade stand business on January 1, 2023. To buy all the required equipment and supplies to get started, you estimate that you will need **\$50k** plus an extra **\$100k**.
2. You open up a business checking account into which you put **\$100k** of your own money. You incorporated and issued yourself, the sole shareholder, **5,000 shares**.
3. In addition, you borrow **\$50K** from the bank at a **10%** annual interest rate.
4. You buy **\$20k** worth of lemons and paper cups (just enough to make **100k cups** of lemonade). You also buy a lemon squeezer for **\$15k**, a lemonade stand for **\$15k**, and a cash register for **\$2k**.
5. You estimate the lemons squeezer and the lemonade stand will have useful lives of **3 years** and the cash register will have a useful life of **5 years**. At the end of the useful life assume the assets will be thrown away (salvage value of \$0).
6. You operate the business for a year and sell **100k cups** of lemonade for **\$1 each**.
7. You hired a lemonade mixer and paid them **\$15k** for the year.
8. In addition, you also hired a cashier and paid them **\$15k** for the year.
9. The tax rate is 40%.

(Feel free to do this in Excel if you find it easier).

\$ In thousands

Jan. 1 - Dec. 31, 2023

Revenue

100.00

Cost of Goods Sold

(20.00)

Gross Profit

80.00

SG&A

(30.00)

D&A

(10.40)

EBIT

39.60

Interest Expense (Net)

(5.00)

Pretax income

34.60

Tax expense

(13.84)

Net Income

20.76

Share Outstanding

5,000

EPS

4.15

EBITDA

50.00

Showing work:

			expense	useful life (y)	amortized cost
Revenue	100,000.00		15,000	3	5000
Cost of Goods Sold	20,000.00		15,000	3	5000
Gross Profit	80,000.00		2,000	5	400
SG&A	30,000.00				
EBIT	39,600.00		worker 1	worker 2	SG&A
Interest			15,000	15,000	30,000

Expense (Net)	5,000.00				
Pretax income	34,600.00				
Tax expense	13,840.00		input	rate	tax/interest
Net Income	20,760.00		50,000	10%	5000
Share Outstanding	5,000		34,600.00	40%	13,840.00
EPS	4.15				
EBITDA	39,600.00		depreciation	amortization	fixed cost
			10400	0	20,000